

COMP 8851 State of the Art Text-to-Speech Technologies and Detecting Voice Cloning Attacks using Automated Speech Assessment

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Abstract. Voice cloning technologies have drastically improved in the past years and has disrupted the voice over industry with its cost and time effective applications. However, its malicious use has also alarmed everyone and undermined the trust of users in digital exchanges. The aim of this project is to review state of the art voice cloning technologies and to investigate counter research on different methods that can detect voice cloning attacks automatically from the original to avoid the malicious use of computer generated speech. The project concludes by identifying the critical areas in literature that have potential to be investigated by researchers.

Keywords: State of the art · Voice cloning technologies · Text-to-speech · Deep fake · Synthetic speech · Mean opinion score · Speech quality · Synthetic speech detection

1 Introduction

1.1 Aims

In this report, we will be reviewing the state-of-the-art technologies for voice cloning. Moreover, we will also be investigating counter research on different methods that can detect computer generated speech automatically from the original to avoid the malicious use of voice cloning. Based on performed research, this report will provide an overview of voice cloning and detection technologies for important stakeholders involving individuals, organizations, IOT based firms, researchers and governments.

1.2 Background

In the past decade, the interchange of media and information has evolved and become dependent on digital exchanges for the delivery of news, entertainment and daily communications. Within this digital realm, computer generated multimedia formats including text, images, videos and speech using artificial intelligence (AI) have also progressed and become more accessible, natural and realistic than ever. Our point of focus in this paper is computer generated speech which is also called synthetic speech

or audio deep fake. With the breakthrough in research and experimentation, synthetic speech became more indistinguishable from bonafide speech. Synthetic speech was initially considered to be uplifting the audio industry by innovating the audio content generation.

However, with the adaptation of these emerging audio generative models, risks have also emerged as these deep fakes are also used by cyber criminals for malicious activities. For example, previously on social media, doctored videos of renowned personalities including Joe Biden, Barack Obama, Joe Rogan, Morgan Freeman and others using AI powered speech have been going viral. Furthermore, the use of cloned voice was also involved in the \$35 Million bank heist making voice clone frauds and malicious activities an alarming factor and major concern for organizations and individuals around the globe (Holloway, 2020). Additionally, these realistic sounding voice cloning technologies have the potential to be used for performing extortion and fraud, framing people, falsifying online identities, falsifying electronic evidence, spreading misinformation, derailing political structures, influencing public opinion, scamming people and so on. Therefore, it has become crucial to identify and manoeuvre the use of deep fakes as the technological advancement is taking place.

1.3 Research Problems and the Context

Recently, traditional machine learning algorithms and exhausting data gathering and labeling approaches were used to train voice cloning technologies also known as text-to-speech(TTS) systems. The use of TTS synthesis has become more realistic as deep learning based models have become an excellent modeler of human voice, the results of the deep learning models have revolutionized the world of both content creators and cybercriminals as discussed above in background. Several deep neural networks and algorithms have the ability to mimic characteristics of a human voice by learning prosody, articulation, tone, fluency, style, cadence, timbre, pitch and even expressions of a speaker. To investigate the advancements in TTS systems and to control the growing incidents of voice cloning frauds, we will be addressing the following research questions:

- What are the state of the art voice cloning technologies?
- What are the different ways in which voice cloning technology can be automatically detected from the original?

1.4 Expected Outcomes

Considering the aforementioned project aims and research questions, following are the two major expected outcomes of this paper:

1. Conduct a comprehensive review of state of the art voice cloning technologies.
2. Identify the different ways in which voice cloning technology can be automatically detected from the original.

1.5 Benefits and Significance

As voice cloning technology is enormously adapted by the content creators and cyber criminals, it has the potential to innovate the voice over industry and at the same time it may exceptionally increase the fraudulent activities in the online world. The findings of this research is aiming to assist researchers in understanding what the most natural sounding voice cloning technologies are and how the quality of computer generated voice can be assessed and further improved. It will also focus on how the researchers can speed up the research and experimentation process using automated assessment methods. It will be mainly focusing on how computer generated voices can be detected automatically using different techniques for avoiding malicious use of cloned voices such as fake voice authentications, spreading fake news, fraud and other malicious activities. This paper will contribute towards exploring synthetic voice detection techniques such as higher order spectral analysis and vocal tract reconstruction which are effective against the state of the art TTS models.

2 Research Methodology

2.1 Methodology

The report is primarily focusing on the literature review and conceptual development of the voice cloning technologies and their detection. The paper will further present a comprehensive review of evolution of synthetic speech, state of the art voice cloning technologies, what makes them state of the art, how their quality is assessed, and how the output of a TTS system can be identified as synthetic automatically.

To address these problems, we will be reviewing existing research and publications to analyze the different ways using which cloned speech can be detected. In the end, we will also be reviewing the effectiveness of discussed ways in existing research and the limitations of these deep fake voice detection methods. We will be comparing multiple TTS models and distinctive ways to detect synthetic audio automatically.

2.2 Timeline

March 8, 2023: Conduct the background research and identify the relevant research articles, documents and weblinks on state of the art voice cloning technologies and detecting voice cloning attacks. Based on initial findings, complete and submit the initial proposal.

March 29, 2023: Determine the primary contributing factors on state of the art cloning technologies and detecting voice cloning attacks. Identifying the potential technological solutions and findings based on existing literature. Revise the initial proposal and submit it.

May 3, 2023: Review and identify the additional contributing factors and technology solutions on both research questions along with automated assessment of the speech quality. Map some of the solutions with the contributing factors and to make the preliminary suggestions for relevant stakeholders. Complete and submit a preliminary report from the revised proposal.

May 23, 2023: Review and finalize the contributing factors and prospective solutions on state of the art cloning technologies and detecting voice cloning attacks. Update and revise the solutions mapping. Finalize all the recommendations for relevant stakeholders and identify the gaps in the existing literature. Prepare and give the final presentation. Complete and submit the final report.

3 Literature Review

In order to detect voice cloning attacks, we need to have an understanding of available voice cloning architectures, technologies and methods for evaluating the quality of these technologies. To identify these technologies and detection methods, several studies have already been performed by researchers. We will examine the progression of voice cloning/speech generation technologies along with quality assessment methods in the initial review. Additionally, we will review automation of speech quality assessment. Furthermore, we will discuss the multi speaker model and state of the art voice cloning technologies. And lastly, we will review the methods using which voice cloning attacks can be detected automatically.

3.1 Speech Generation and Quality Assessment

3.1.1 Progression of Speech Generative Technologies

A research review conducted by Karagiannakos (2021) shows the sequential development of the voice cloning technologies. He describes the advent of synthetic speech began with the concatenation synthesis which used a concatenation technique on pre-recorded speech segments to generate speech. Concatenation synthesis had natural quality however its output contained audible glitches. Then it progressed to statistical parametric synthesis which extracts characteristics of audio signals such as frequency spectrum forming up the vocal tract, fundamental frequency and duration of speech segments. It overcame the need for an audio database used in previous concatenation synthesis methods. Nonetheless, its quality was still not ideal. To examine the state of the art technologies, he described the evaluation of speech

synthesis based on mean opinion score(MOS) which is often used to evaluate the quality of synthesized speech. MOS is the average of the people's opinions who are listening and evaluating the speech. It is a subjective approach as it is influenced by the assessor's personal feelings and opinion about the speech. MOS scale has a range between 0(lowest) to 5(highest) where the range of real human speech lies between 4.3 to 4.8. Based on the MOS of parametric synthesis, he further stated how the problem of bad quality is then covered by deep learning based speech synthesis. His studies further reveal that there are several architectures discovered and experimented till now including the Inverse Autoregressive Flows (IAFs), transformers as teacher-student distillation, Generative adversarial networks (GANs), Variational Autoencoders (VAEs), flow-based TTS, etc. which possess different approaches, training/inference times and qualities. Table 1 shows the MOS of these deep learning based models along with some of their features and limitations.

3.1.2 Speech Quality Metrics

Hinterleitner (2017) research shows that five commonly used perceptual quality metrics derived for speech are (1) naturalness of voice (2) prosodic quality (3) fluency and intelligibility (4) absence of disturbances, and (5) calmness. These metrics can be used to do objective evaluation of speech by training neural network based deep learning models. Other metrics mentioned by Chan et al. (2016) are word error rate or phoneme error rate, acoustic features such as pitch, volume, and speech rate, fluency, articulation meaning clarity of spoken phonemes also determine the quality of speech. Training a deep learning model having labeled data by considering above metrics can be used to design a system for automated assessment of speech.

3.1.3 Automatic Speech Quality Assessment

Salza et al. (1996) discuss that speech is assessed subjectively by using techniques such as MOS and paired comparison tests. These methods are challenging and time consuming as they are subjective methods of evaluation performed by humans. With the research and development of several TTS systems, the need for assessing the quality of these systems during experiments has increased drastically. Therefore, building automated speech processing systems that can evaluate the quality of these systems also becomes vital for speeding up the experimentation and research process. Primarily, these systems are also beneficial for addressing the problem of deep fake detection.

In a study, Jaiswal et al. (2019) research proposes a method to automatically compute the quality of speech being close to human by using speech evaluation metrics and Virtual Mean Opinion Score (vMOS) of the speech under assessment or generated by TTS models. The proposed method uses an end-to-end neural network to calculate vMOS which is qualitatively an alternative of manually obtained MOS. The approach becomes objective as it is evaluated without human intervention and it is not influenced by the assessor's personal feelings and opinion. The aim of this approach is to predict

only the evaluation score, not the speech. The authors showed that the approach therefore is faster and needs a small dataset to train the model. The method was tested to examine the speech produced by the state of art TTS models. The model developed by authors produced an output which lies on a scale from 0 to 1. They observed vMOS of 0.6675, 0.4945 and 0.4890 for Wavenet2, Tacotron and Deepvoice3 respectively where human speech was 0.6682. These vMos scores predicted by the neural network corresponded with the scale of manually calculated MOS scores which shows its effectiveness and reliability as compared to manual assessment process.

3.1.4 Multi Speaker Model

A study conducted by Jia et al. (2018) indicated the architecture used in deep learning models. The paper presents TTS synthesis based on a neural network system that has the capability to generate speech in the voice of multiple speakers, including those unseen during training. His findings show that text is passed as input to acoustic feature generators. These models generally have three components (1) Speaker encoder network having multiple speaker embeddings containing speaker characteristics (2) Sequence-to-sequence synthesis network based on Tacotron 2, which generates mel spectrogram using input text and speaker embedding (3) neural vocoder which transforms the mel spectrogram to waveform signal. The authors of this paper demonstrate that the proposed model has the ability to synthesize natural sounding speech of speakers that are not in the training data. This is achieved by transferring the knowledge of speaker variability learned by the discriminatively trained speaker encoder. Their study shows that the model can also have best generalization performance by training the speaker encoder on a diverse and large speaker set. They also demonstrate that the model has learned the high quality speaker representation by utilizing the randomly sampled speaker embeddings to generate speech in the voices of new speakers, which are different from those used for training datasets.

3.1.5 State of the Art Voice Cloning Technologies

We discussed in the previous sections the progression of synthetic speech, assessment of speech quality and multispeaker abilities. On further review of the existing publications, it is found that there are multiple factors that contribute towards the state of the art voice cloning technologies. These factors involve (1) Human-ness, how natural model's output sounds and how close it is to real speech, some models sound robotic (2) Inference time, how much time a model takes to convert the text to audio making it effective in real time applications (3) End-to-end systems, ability to handle both components: synthesizer and vocoder (4) Control on features during inference such as style, speed, expressiveness, etc. (5) Speech features including correctness in pronunciation, prosody, intonation and emphasis on words (6) Multilingual and multi speaker capabilities, ability to train multiple speakers in one training and handling multiple languages (7) Training dataset, some models require huge datasets for training which makes them expensive for training and experiment, modern architectures needs

only 10 to 20 seconds of data to learn a new speaker's style (8) Deep learning architectures, the ability of a model to learn complex relationship between text inputs and corresponding speech output, etc.

When comparing the performance of different methods, several factors are considered. Some methods exhibit high costs but offer superior quality. Conversely, other methods excel in real-time performance but may yield speech that sounds artificial or machine-like. However, based on MOS, we can say that Tacotron 2 and transformer based TTS systems outperform other networks. In table 1, we can compare the real human speech with other speech generation methods. It can be observed that the models at the end are latest and state of the art. For example, GANs based TTS, transformer based TTS systems are best performing models as they have faster audio generation and have the ability to control style in a speech. In further review, we will be reviewing the methods which can detect speech generated by these models.

Table 1. Comparison of human speech with voice cloning technologies

Methods	Features and Limitations	MOS
Real Human Speech	Represents the natural speech of a human	4.3 - 4.8
Wavenet	Very slow sample prediction, computationally expensive inference	4.21
Deep Voice	Achieved real-time inference, predicts phoneme duration, costly to train the model	2.67
Tacotron	End-to-end, adapts to new data easily, robust, costly to train the model	3.82
Deep Voice 2	Enhancements to tacotron for supporting multispeaker and attention mechanism, costly to train model	3.53
Deep Voice 3	Does not generalize well for unseen samples	3.78
Parallel WaveNet	1000x faster than other previous networks, does not generalize well for unseen samples	4.41
Tacotron 2	Impressive results, lack of real time speech synthesis	4.53
Global Style Tokens	Control over speech style	4.0
Transformers TTS	Handles low efficiency during training and inference, handles difficulty in modeling long dependencies using RNNs	4.39
Fast Speech 2	38x faster, the teacher-student distillation pipeline is complicated	3.62
GAN based TTS	Faster audio generation than previous systems, GANs gives stylistic control	4.08

3.2 Synthetic Speech Detection

According to Masood et al., (2023), audio deepfakes are classified in two categories (1) text-to-speech synthesis and (2) voice conversion. Conversely, voice replay attacks are

not categorized as synthetic, as they involve the recording of target speaker's voice. In the forthcoming literature review, our research focuses on deepfake detection only.

In further review for detecting the voice cloning attacks, a study of Malik (2019) describes a framework to detect cloned speech attacks on voice-driven interfaces and services using higher order spectral analysis. He also discussed that researchers have proposed to detect deep fake by capturing the artifacts in the output signal of the synthetic audio which are not observable in the natural speech. Artifacts can be captured to differentiate between cloned and bona fide speech by using quadrature phase coupling (QPC) in the estimated bicoherence, Gaussianity test statistics, and linearity test statistics. His paper showed the detection is almost close to a perfect classification of real and synthetic audio. The only setback is that it is trained on huge labeled datasets and may not generalize on test datasets. Innovative approaches are emerging in this area by utilizing the high order spectral analysis in order to extract artifacts from generative model cloned speech and also provide promising directions for future research.

In another study, Blue et al. (2022) focused on detection of audio deep fakes through vocal tract reconstruction. He along with other researcher colleagues has developed a novel mechanism using techniques from the field of articulatory phonetics. In which, they applied fluid dynamics to estimate the arrangement of the human vocal tract during speech generation. They also estimated the vocal tract of computer generated speech which forms the shape of a drinking straw that forms impossible anatomical arrangements. The comparison of these estimates can clearly predict the real and fake speech. Their mechanism uses a deep fake detector which is able to detect deep fake audio from a huge data set with 99.5% recall and 99.9% precision. The paper discusses the limitations of this approach and how biologically restricted aspects of human speech generation can be a powerful tool for the detection of audio deep fakes. The main significance of this approach is that it can detect synthesized speech of multiple technologies and it does not need a huge training dataset as compared to the approach followed in the previously mentioned approach.

4 Outcomes and Discussion

Based on the literature review, we have observed that the voice cloning technology has evolved significantly in recent years. Synthetic speech has progressed from concatenation synthesis to statistical parametric synthesis, and in the past decade, to deep learning based speech synthesis. The quality of the synthesized speech has improved so much making it more difficult to distinguish synthetic speech from natural speech. The deep learning-based speech synthesis models also have the ability to generate speech in the voice of different speakers. Voice cloning technologies have also improved in terms of training/inference time. Another area of research we have explored is to evaluate the output of these synthesis models using MOS which is essential for both development of state of the art models and their detection. Moreover,

we have also found that audio processing systems such as vMOS prediction can automate and speed up the research and experimentation process.

The evolution of voice cloning technology poses a significant threat to the security of voice-driven interfaces, services and systems. To address these threats, we also reviewed the research taking place in methods for detecting voice cloning attacks, such as high order spectral analysis and vocal tract reconstruction. Currently, state of the art speech generation methods are unable to replicate vocal tract formation of actual human speakers. Consequently, vocal tract reconstruction becomes a state of the art deep fake detection approach, as it possesses the capacity to detect all existing voice cloning technologies. Nevertheless, this detection approach can be deceived in the future if deep learning models are enhanced to accurately replicate the vocal tract characteristics of real speaker.

Other techniques, such as anomaly and artifacts detection in audio have also demonstrated effectiveness in the majority of detection cases. However, these methods may fail to detect deep fakes produced by speech generation technologies that produce quality output and have fewer detectable artifacts. Automated assessment of vMOS can also classify the deep fakes with low vMOS score. These techniques used in hybrid approaches with automated assessment of speech may provide promising directions for future research in this field.

5 Conclusion and Future Outlook

To recapitulate, voice cloning technologies have advanced significantly in recent years, making synthetic speech indistinguishable from human speech. While these technologies have the potential to revolutionize the voice over industry, they also pose significant risks, including fraudulent activities and the spread of misinformation. In this paper, we have reviewed the state-of-the-art voice cloning technologies and investigated methods for automatically detecting computer-generated speech. Our findings provide insights into the advancements in voice cloning technologies, such as deep learning-based models such as transformers, which showed superior performance in terms of speech quality. Models such as fast speech 2 has drastically improved the training and inference time. We have also reviewed results, effectiveness and limitations of some approaches. Finally, the paper explores techniques for automatically detecting synthetic speech, including the use of speech evaluation metrics and virtual Mean Opinion Score (vMOS). The research contributes towards the understanding of natural-sounding voice cloning technologies and provides recommendations for stakeholders to mitigate the malicious use of cloned voices. It also highlights the importance of developing effective detection methods to combat deep fake voice fraud and protect individuals and organizations from potential harm.

For the future work, accurate detection of deep fake attacks in real time is a potential research area that should be addressed by researchers. They should explore the integration of hybrid approaches combining automated speech assessment and deep fake detection methods. Researchers should also investigate how the deep fake

classification systems can be integrated with audio-driven systems in the real-time to avoid scams. One area of focus could be developing a mobile application that captures the voice of the caller and then processes it in the background using a vocal tract detection system which evaluates the vocal tract formation of the caller. It should alarm the user if a computer generated voice is detected. Another focus area is to integrate the synthetic speech detection system into existing communication applications. For example, voice calling apps can include an additional layer of real-time speech analysis that detects and alerts users about potential voice cloning attempts.

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